

EXP1:

Variable Transformation between Cartesian, Cylindrical and Spherical coordinate system to obtain the location of any object.

TEST CASE 1

Code:

```
1 clc;
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z='); r=sqrt(x^2+y^2);
5 pi=atan2(y/x);
6 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=', [r pi z]);
7 r=sqrt(x^2+y^2+z^2);
8 theta=acosd(z/r); pi=atan2(y/x);
9 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=', [r theta pi]);
10
```

Calculation:

ECAD504 Lab observation

1) ~~Example~~ case: 1 $x = -1$
 $y = 5$
 $z = -8$

Rectangular

Cylindrical

$$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (5)^2} = \sqrt{1 + 25} = \sqrt{26} = 5.099$$
$$\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$$
$$z = -8$$

Spherical

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(-1)^2 + (5)^2 + (-8)^2} = \sqrt{1 + 25 + 64} = \sqrt{90} = 9.48$$
$$\theta = \cos^{-1}\left[\frac{z}{r}\right] = \cos^{-1}\left[\frac{-8}{9.48}\right] = \cos^{-1}(-0.844) = 147.48^\circ$$
$$\phi = \tan^{-1}\left[\frac{y}{x}\right] = \tan^{-1}\left[\frac{5}{-1}\right] = -78.69^\circ$$

OUTPUT:

Scilab 2025.0.0 Console

File Edit Control Applications ?



Scilab 2025.0.0 Console

Enter the value of X=-1

Enter the value of y=5

Enter the value of z=-8

"Cartesian to cylindrical coordinate system of p(r,pi,z)="
5.0990195 -78.690068 -8.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
9.486833 147.48748 -78.690068

-->

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

File Edit Format Options Window Execute ?



exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

exp333.sci X

```
1 clc;-
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atan2(y/x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]');
8 r=sqrt(x^2+y^2+z^2);
9 theta=acosd(z/r);
10 pi=atan2(y/x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]');
12
```

TEST CASE 2

Code:

```
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 z=input('Enter the value of z=');
5 x=r*cosd(pi);
6 y=r*sind(pi);
7 z=z;
8 disp('cylindrical to cartesian coordinate system of p(x,y,z)=',[x y z]);
9 r=sqrt(r^2+z^2);
10 pi=pi; theta=acosd(z/r);
11 disp('cylindrical to spherical coordinate system of s(r,pi,theta)=',[r pi theta]);
12
```

Calculation:

2) case: 2

$\phi = 5$

$\phi = 30^\circ$

$z = -5$

cylindrical

Rectangular

$x = \rho \cos \phi = 5 \cos(30^\circ) = 4.33$

$y = \rho \sin \phi = 5 \sin(30^\circ) = 2.5$

$z = z = -5$

Spherical

$\rho = \sqrt{\rho^2 + z^2} = \sqrt{(5)^2 + (-5)^2} = \sqrt{25 + 25} = \sqrt{50} = 7.07$

$\phi = \phi = 30^\circ$

$\theta = \tan^{-1} \left[\frac{\rho}{z} \right] = \tan^{-1} \left[\frac{5}{-5} \right] = \tan^{-1}(-1) = \pm 45^\circ$

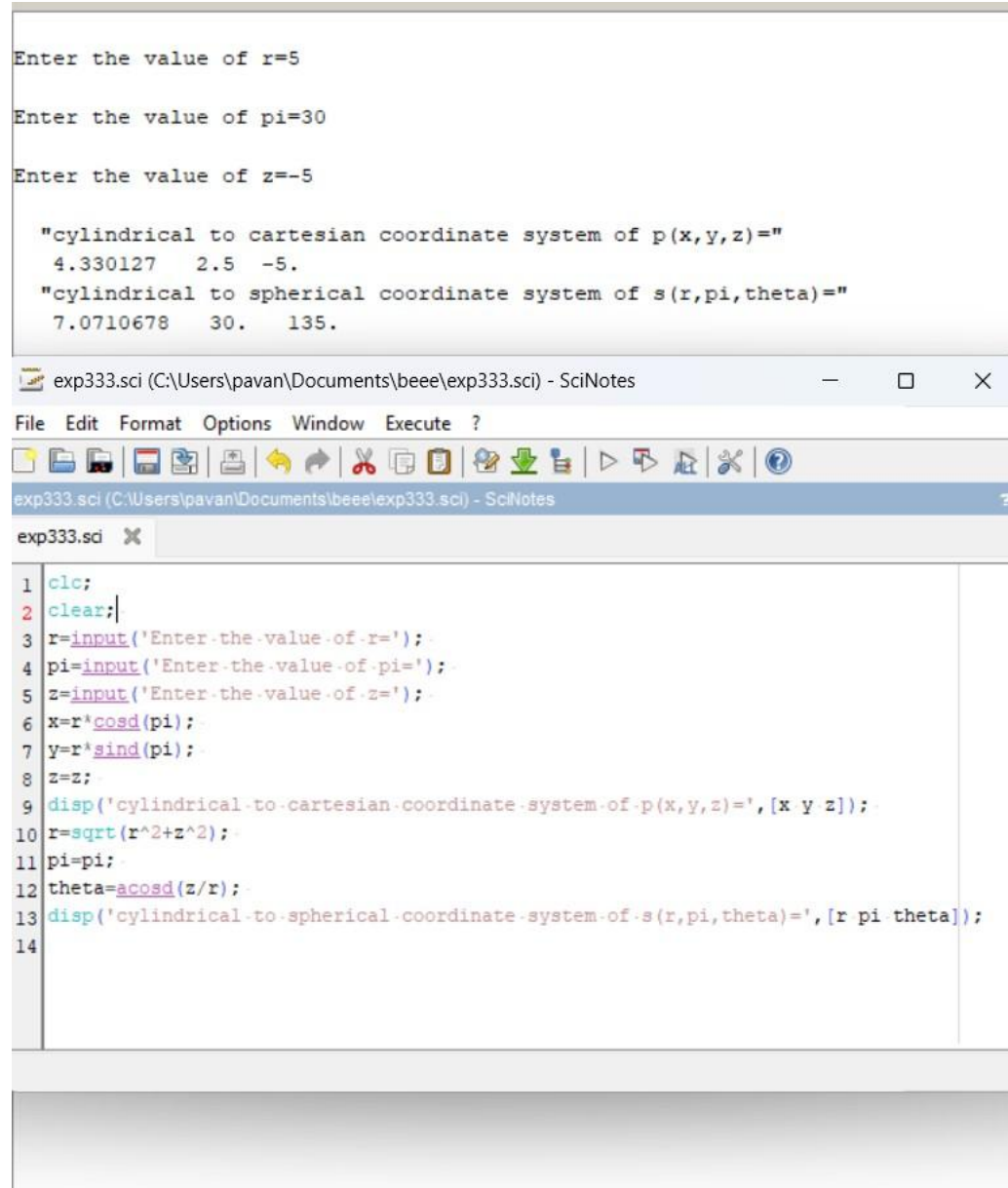
Output:

```
Enter the value of r=5

Enter the value of pi=30

Enter the value of z=-5

"cylindrical to cartesian coordinate system of p(x,y,z)="
4.330127 2.5 -5.
"cylindrical to spherical coordinate system of s(r,pi,theta)="
7.0710678 30. 135.
```



The screenshot shows a SciNotes window titled "exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes". The window contains MATLAB code that prompts the user for values of r, pi, and z, then calculates and displays the conversion from cylindrical to Cartesian and spherical coordinate systems. The code is as follows:

```
1 clc;
2 clear;
3 r=input('Enter the value of r=');
4 pi=input('Enter the value of pi=');
5 z=input('Enter the value of z=');
6 x=r*cosd(pi);
7 y=r*sind(pi);
8 z=z;
9 disp('cylindrical to cartesian coordinate system of p(x,y,z)=[x y z]');
10 r=sqrt(r^2+z^2);
11 pi=pi;
12 theta=acosd(z/r);
13 disp('cylindrical to spherical coordinate system of s(r,pi,theta)=[r pi theta]');
14
```

TEST CASE 3

Code:

```
1  clc;
2  r=input('Enter the value of r='); pi=input('Enter the value of pi=');
3  theta=input('Enter the value of theta=');
4  x=r*sind(theta)*cos(pi);
5  y=r*sind(theta)*sin(pi);
6  z=r*cosd(theta);
7  disp('spherical to cartesian coordinate system of p(x,y,z)=', [x y z]);
8  p=r*sin(theta);
9  pi=pi; z=r*cos(pi);
10 disp('spherical to cylindrical coordinate system of s(p,pi,z)=', [p pi z]);
11
```

Calculation:

3) Case: 3

$r = 3$
 $\theta = 60^\circ$
 $\phi = 120^\circ$

Spherical

Rectangular

$$x = r \sin \theta \cos \phi = 3 \sin(60^\circ) \cos(120^\circ) = -2.115$$
$$y = r \sin \theta \sin \phi = 3 \sin(60^\circ) \sin(120^\circ) = 2.598$$
$$z = r \cos \theta = 3 \cos(60^\circ) = 1.5$$

Cylindrical

$$\rho = r \sin \theta = 3 \sin(60^\circ) = 2.598$$
$$\phi = \phi = 120^\circ$$
$$z = r \cos \theta = 3 \cos(60^\circ) = 1.5$$

Output:

Enter the value of r=3

Enter the value of pi=60

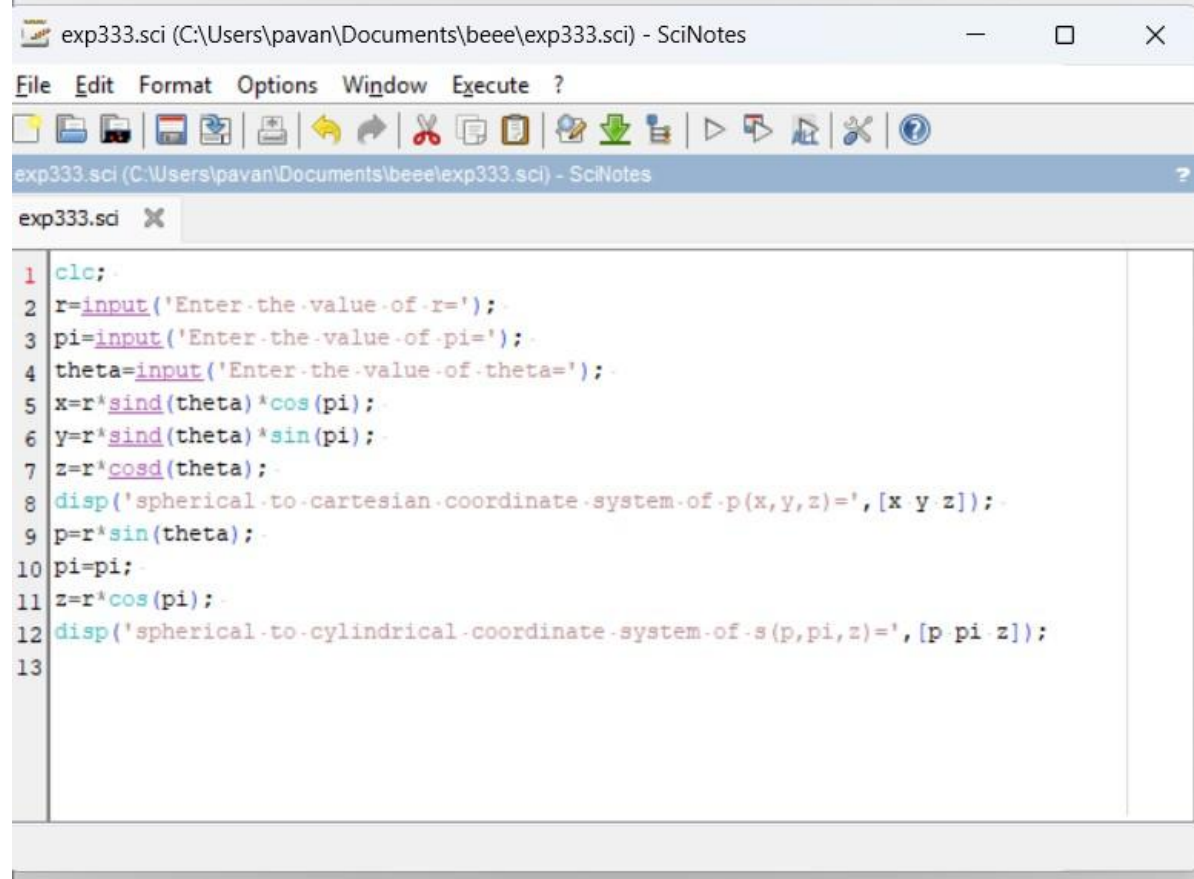
Enter the value of theta=120

"spherical to cartesian coordinate system of p(x,y,z)="

-2.4744415 -0.7919212 -1.5

"spherical to cylindrical coordinate system of s(p,pi,z)="

1.7418336 60. -2.8572389



The screenshot shows a SciNotes window titled "exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes". The window has a menu bar with "File", "Edit", "Format", "Options", "Window", "Execute", and "?". Below the menu bar is a toolbar with various icons for file operations and execution. The main text area contains the following MATLAB code:

```
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 theta=input('Enter the value of theta=');
5 x=r*sind(theta)*cos(pi);
6 y=r*sind(theta)*sin(pi);
7 z=r*cosd(theta);
8 disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z]');
9 p=r*sin(theta);
10 pi=pi;
11 z=r*cos(pi);
12 disp('spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z]');
13
```